

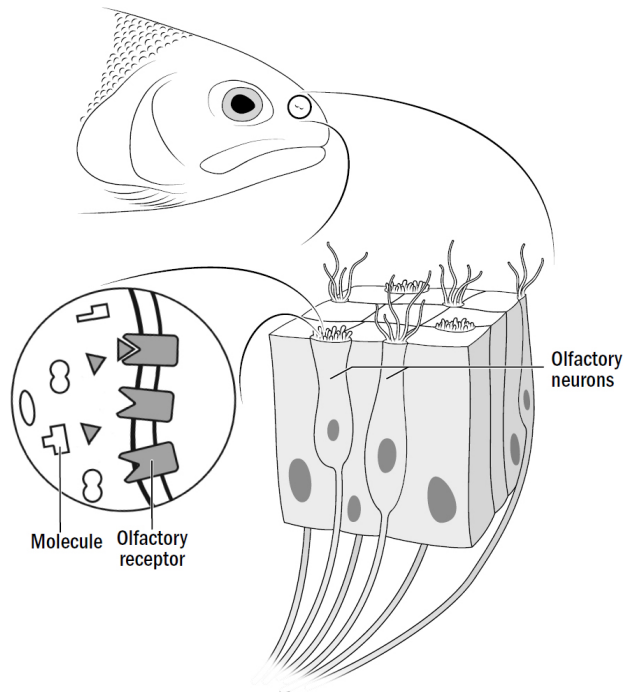


Figure 7.1: Inside the nose of a vertebrate

Original art by Rebecca Gelernter

However, most of the information about the world around you can't be found in a single activated neuron but only in the pattern of activated neurons. You can distinguish a car from a house based on the pattern of photons hitting your retina. You can distinguish the ramblings of a person from the roar of a panther based on the pattern of sound waves hitting your inner ear. And, yes, you can distinguish the smell of a rose from the smell of chicken based on the pattern of olfactory neurons activated in your nose. For hundreds of millions of years, animals were deprived of this skill, stuck in a perceptual prison.

When you recognize that a plate is too hot or a needle too sharp, you are recognizing attributes of the world the way early bilaterians did, with the activations of individual neurons. However, when you recognize a smell, a face, or a sound, you are recognizing things in the world in a way that was beyond early bilaterians; you are using a skill that emerged later in early vertebrates.

Early vertebrates could recognize things using brain structures that decoded *patterns* of neurons. This dramatically expanded the scope of what animals could perceive. Within the small mosaic of only fifty types of

olfactory neurons lived a universe of different patterns that could be recognized. Fifty cells can represent over *one hundred trillion patterns*.*

HOW EARLY BILATERIANS RECOGNIZED THINGS IN THE WORLD	HOW EARLY VERTEBRATES RECOGNIZED THINGS IN THE WORLD
A single neuron detects a specific thing	Brain decodes the pattern of activated neurons to recognize a specific thing
Small number of things can be recognized	Large number of things can be recognized
New things can be recognized only through evolutionary tinkering (new sensory machinery needed)	New things can be recognized without evolutionary tinkering but through learning to recognize a new pattern (no new sensory machinery needed)

Pattern recognition is hard. Many animals alive today, even after another half billion years of evolution, never acquired this ability—the nematodes and flatworms of today show no evidence of pattern recognition.

There were two computational challenges the vertebrate brain needed to solve to recognize patterns. In [figure 7.2](#), you can see an example of three fictional smell patterns: one for a dangerous predator, one for yummy food, and one for an attractive mate. Perhaps you can see from this figure why pattern recognition won't be easy—these patterns overlap with each other despite having different meanings. One should trigger escape and the others approach. This was the first problem of pattern recognition, that of *discrimination*: how to recognize overlapping patterns as distinct.

The first time a fish experiences fear in the presence of a novel predator smell, it will remember that specific smell pattern. But the next time the fish encounters that same predator smell, it won't activate the exact same pattern of olfactory neurons. The balance of molecules will never be identical—the age of the new arthropod, or its sex, or its diet, or many other things might be different that could slightly alter its scent. Even the background smells from the surrounding environment might be different, interfering in slightly different ways. The result of all these minor perturbations is that the next encounter will be *similar* but *not the same*. In [figure 7.3](#) you can see three examples of the olfactory patterns that the next encounter with the predator smell might activate. This is the second challenge of pattern recognition: how to *generalize* a previous pattern to recognize novel patterns that are similar but not the same.

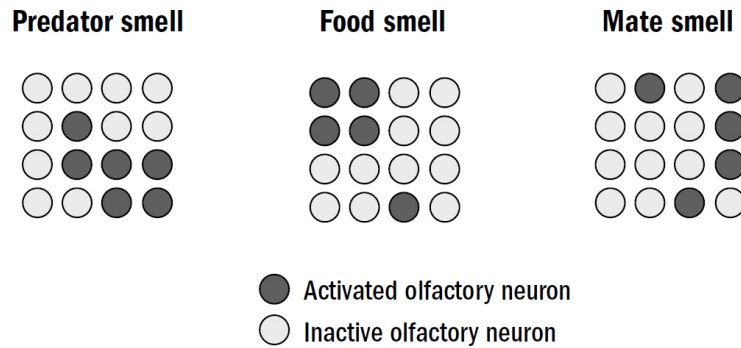


Figure 7.2: The discrimination problem

Figure by Max Bennett

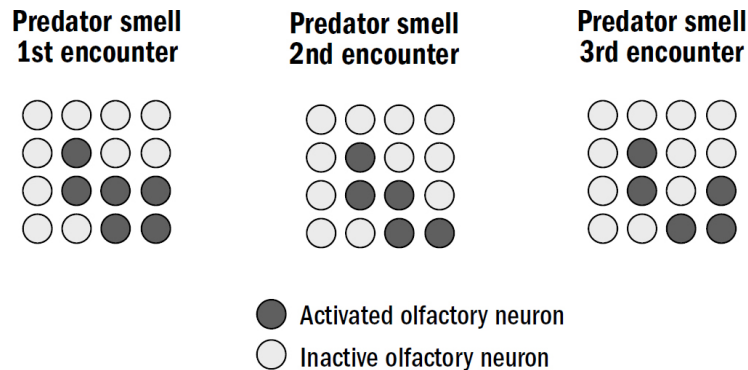


Figure 7.3: The generalization problem

Figure by Max Bennett

How Computers Recognize Patterns

You can unlock your iPhone with your face. Doing this requires your phone to solve the generalization and discrimination problems. Your iPhone needs to be able to tell the difference between your face and other people's faces, despite the fact that faces have overlapping features (discrimination). And your iPhone needs to identify your face despite changes in shading, angle, facial hair, and more (generalization). Clearly, modern AI systems successfully navigate these two challenges of pattern recognition. How?

The standard approach is the following: Create a network of neurons like in [figure 7.4](#) where you provide an input pattern on one side that flows through layers of neurons until they are transformed into an output on the other end of the network. By adjusting the weights of the connections